

BMO: Benchmarking Model-quantized Offline Agents for Sustainable Language Tutoring

A Sovereign Edge AI Framework for Resource-Constrained Pedagogical Environments

Abstract

Current AI language learning tools rely heavily on cloud APIs, imposing a hidden carbon footprint. This paper introduces BMO, a fully autonomous, voice-to-voice French language learning companion running entirely offline on entry-level edge hardware. By coupling Whisper ASR, 4-bit GGUF SLMs, and Kokoro-82M TTS, we evaluate the trade-offs between quantization and pedagogical accuracy. Our empirical telemetry demonstrates a 66.0% pass rate while consuming only 27.49 Wh of energy.

1. Introduction & Core Benchmark Findings

BMO evaluates small language models (SLMs) running locally on CPU-only hardware. Through a 200-question standardized exam of A2/B1 French conversational prompts, we measured pedagogical error correction, latency, and energy telemetry. Key findings include:

- Pedagogical Pass Rate: 66.0% (132/200 turns passed)
- Total Energy Consumption: 27.49 Wh (0.137 Wh/sentence)
- Total Carbon Footprint: 19.61 g CO₂e (0.098 g CO₂e/sentence)
- Average Response Latency: 13.79 seconds per turn on 4-thread CPU

2. Edge Architecture & State Engines

BMO couples three specialized state engines:

1. RoleplayEngine: Handles 3-4 turn real-life scenario simulations (Parisian Cafe, Bakery, Train Station, Hotel) with automated exit debriefs.
2. ScaffoldingEngine: Implements a 3-tiered hint hierarchy and hesitation recovery playing at slowed 0.70x speech pacing.
3. SessionMemoryEngine: Maintains longitudinal student memory in session_review.json, generating startup warm-up quizzes and extracting vocabulary on session exit.